



s2xin

Reference Manual

Version 1.10

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1 Overview

1.10

- **Changed**
 - s2xinDemodulator from 1.7 to 1.8;
 - bissDescrambling from 1.5 to 1.6;
 - bissTypes from 1.2 to 1.3;
 - bissVersion1 from 1.2 to 1.3;
 - bissVersion2 from 1.5 to 1.6;

1.9

- **Changed**
 - component from 1.4 to 1.5;
 - tsstatus from 1.3 to 1.5;
 - s2xinStatus from 1.5 to 1.6;

1.8

- **Changed**
 - s2xinDemodulator from 1.6 to 1.7;
 - bissDescrambling from 1.4 to 1.5;
 - bissVersion2 from 1.4 to 1.5;

1.7

- **Added**
 - bissCaSettings version 1.0;
- **Changed**
 - s2xinDemodulator from 1.5 to 1.6;
 - s2xinStatus from 1.4 to 1.5;
 - bissDescrambling from 1.3 to 1.4;
 - bissVersion2 from 1.3 to 1.4;
 - component from 1.1 to 1.4;
 - tsstatus from 1.1 to 1.3;

2 bissCaSettings (1.0)

2.1 Type Reference

2.1.1 BissCaEcmInterval

BISS CA ECM Repetition Interval

Predefined set of possible ECM repetition interval values, in milliseconds.

Values lower than 100 ms are outside of the BISS CA specification. These values may still be used if the parameter `extendedOptions` is set to true in the `BissCaSettings` struct.

enum

I6	5.7 ms (Value outside BISS CA spec)
I15	14.7 ms (Value outside BISS CA spec)
I50	50 ms (Value outside BISS CA spec)
I100	100 ms
I125	125 ms
I150	150 ms
I175	175 ms
I200	200 ms
I250	250 ms
I300	300 ms

2.1.2 BissCaEntitlementFlags

BISS CA Entitlement Flags

The entitlement flags are parameters for how the entitled receivers shall behave. These values are mandatory part of the stream, and is inserted into the session data, which is the encrypted part of the EMM table.

struct

<code>preventDescrambledForward</code>	bool Prevent descrambling forward
<code>preventDecodedForward</code>	bool Prevent decode forward
<code>insertWatermark</code>	bool Insert watermark

2.1.3 BissCaSettings

BISS CA Specific Settings

struct

<code>cryptoPeriod</code>	int Crypto period (in seconds).
<code>emmBitrate</code>	int EMM bitrate (kbps).
	BissCaEcmInterval

ecmInterval	ECM repetition interval. Note that changes to the repetition interval will not take effect until the next crypto period (the next time the ECM content is updated).
emmPreferredPid	int EMM preferred PID value. If the PID value is not available, the first larger PID value available will be used instead. If no larger PID value is available, then the first smaller available value will be used.
ecmPreferredPid	int ECM preferred PID value. If the PID value is not available, the first larger PID value available will be used instead. If no larger PID value is available, then the first smaller available value will be used.
entitlementFlags	BissCaEntitlementFlags BISS CA entitlement flags. Controls how the entitled receivers shall behave.
extendedOptions	bool Allow for extended options. This means allowing violation of the BISS CA specification. If set to true, then the following requirements of the BISS CA specification are <i>not</i> enforced: - EMM repetition interval must be at least 200 ms - ECM repetition interval must be at least 100 ms Note that the requirement that EMM interval must be larger than the ECM interval is still enforced.

3 bissData (1.3)

3.1 Command Reference

3.1.1 GetMetaData

Get Meta data

The BISS data is write-only. This getter therefore returns a map of Meta data objects in the database.

- message **GetMetaData.Request**
- message **GetMetaData.Response**

3.1.2 UpdateLabel

Update user label for existing BissData

UUIDs that does not exist already in the database will be discarded.

- message **UpdateLabel.Request**
- message **UpdateLabel.Response**

3.1.3 SetBissData

Set BISS data

The BISS data is write-only. This setter gives both data content and meta data. Whereas only the meta data can be retrieved back through the API later.

- message **SetBissData.Request**
- message **SetBissData.Response**

3.1.4 DeleteBissData

Delete BISS data

- message **DeleteBissData.Request**
- message **DeleteBissData.Response**

3.1.5 GetEncryptedSessionWords

Get Encrypted Session Words for all valid session Ids

Returns a list containing the encrypted version of the entered Session Word for each valid Session Id.

- message **GetEncryptedSessionWords.Request**
- message **GetEncryptedSessionWords.Response**

3.1.6 GetRsaPublicKeys

Get RSA public key

Returns a list of RSA public keys if present and valid

- message **GetRsaPublicKeys.Request**
- message **GetRsaPublicKeys.Response**

3.1.7 GetEkIds

Get Entitlement Key Ids (EKIDs)

Returns a list of EKIDs if present and valid

- message **GetEkIds.Request**
- message **GetEkIds.Response**

3.1.8 GetRsaPublicKeyFromBuriedKey

Get RSA public key from buried RSA key

Returns empty optional in case of error.

Only applicable for descrambling modules.

- message **GetRsaPublicKeyFromBuriedKey.Request**
- message **GetRsaPublicKeyFromBuriedKey.Response**

3.1.9 GetEkIdFromBuriedKey

Get Entitlement Key Ids (EKIDs) from the buried RSA key

Returns empty optional in case of error.

Only applicable for descrambling modules.

- message **GetEkIdFromBuriedKey.Request**
- message **GetEkIdFromBuriedKey.Response**

3.2 Type Reference

3.2.1 BissData

Biss data

Write-only data for use in BISS scrambling / descrambling. The data content is entered in clear. Only the meta data is accessible through the API afterwards. This is not the case for the content itself.

struct

meta	BissDataMeta Meta data for the BISS data.
data	string Data content. Format of data is determined by the type.

3.2.2 BissDataMeta

Meta data for BISS data

struct

label	string Label of user data
dataType	BissDataType Type of BISS data

3.2.3 BissDataType

BISS data type

Data type describing the content in a BISS data object

enum

BISS1E_ID	BISS version 1 identifier, 56-bit as hex string
BISS2E_ID	BISS version 2 identifier, 128-bit as hex string
BISS1E_SW	BISS version 1 session word, 48-bit as hex string
BISS2E_SW	BISS version 2 session word, 128-bit as hex string
BISS1E_ENC_SW	BISS version 1 encrypted session word, 64-bit as hex string
BISS2E_ENC_SW	BISS version 2 encrypted session word, 128-bit as hex string
BISS2CA_RSA_PRIV	BISS version 2 RSA private key
BISS2CA_RSA_PUB	BISS version 2 RSA public key

3.2.4 DeleteBissData.Request

struct

ids [list of UUID](#)

3.2.5 DeleteBissData.Response

empty [struct](#)

3.2.6 EncryptesSessionWordRequest

Request object of GetEncryptedSessionWords

struct

sessionWordId [UUID](#)
Id of session word

3.2.7 EncryptesSessionWordResponse

Response object of GetEncryptedSessionWords

The sessionId is the UUIS of the BissData session ID, and the encryptedSessionWord is the corresponding encrypted session word as a hexadecimal string.

struct

data [list of EncryptesSessionWordResponse.data.item](#)

3.2.8 EncryptesSessionWordResponse.data.item

struct

sessionId [UUID](#)
encryptedSessionWord [string](#)

3.2.9 GetEkIdFromBuriedKey.Request

empty **struct**

3.2.10 GetEkIdFromBuriedKey.Response

struct

data **optional string**

3.2.11 GetEkIds.Request

struct

ids **optional list of UUID**

3.2.12 GetEkIds.Response

struct

data **map from UUID to string**

3.2.13 GetEncryptedSessionWords.Request

EncryptesSessionWordRequest

3.2.14 GetEncryptedSessionWords.Response

EncryptesSessionWordResponse

3.2.15 GetMetaData.Request

empty **struct**

3.2.16 GetMetaData.Response

struct

data **map from UUID to BissDataMeta**

3.2.17 GetRsaPublicKeyFromBuriedKey.Request

empty **struct**

3.2.18 GetRsaPublicKeyFromBuriedKey.Response

struct

data **optional string**

3.2.19 GetRsaPublicKeys.Request

struct

ids optional list of UUID

3.2.20 GetRsaPublicKeys.Response

struct

data map from UUID to string

3.2.21 SetBissData.Request

struct

data map from UUID to BissData

3.2.22 SetBissData.Response

empty struct

3.2.23 UpdateLabel.Request

struct

data map from UUID to string

3.2.24 UpdateLabel.Response

empty struct

4 bissDescrambling (1.6)

4.1 Type Reference

4.1.1 DescramblerSettings

- Descrambler Settings
-
- Settings defining how to descramble the transport stream.
- By default, there is a single descrambling settings parameter, where these
- settings are applied to all services in the transport stream (TS).
-
-
-

struct

enabled	bool Descrambling on/off.
	•

method	DescramblingMethod Descrambling settings.
--------	---

4.1.2 DescramblingMethod

DescramblingMethod

variant

fixedKeyRaw	bissTypes.FixedKeyRawDescrambling Fixed key decryption.
-------------	---

biss1	bissVersion1.Descrambling BISS version 1 decryption.
-------	--

biss2	bissVersion2.Descrambling BISS version 2 decryption.
-------	--

5 bissHelpers (1.0)

6 bissSettings (1.0)

6.1 Type Reference

6.1.1 AllowedExtraSettings

Allowed Num of Extra Scrambling Settings

enum

EXTRA_15_ECMS	15 extra settings
EXTRA_7_ECMS	7 extra settings
EXTRA_3_ECMS	3 extra settings

6.1.2 AllowedMode

Allowed (De)Scrambling modes

enum

ALL	CSA and AES algorithms allowed
AES_ONLY	Only AES allowed
CSA_ONLY	Only CSA allowed

6.1.3 BissModuleSettings

BISS Module Setup for a given product (not configurable)

struct

allowedMode	AllowedMode Algorithms supported by product
numExtraPerOutput	AllowedExtraSettings Number of extra settings allowed per output
maxNumOutputs	int Maximum number of outputs
totalEmmBitrateLimit	optional int Set an upper limit for the total EMM playout bitrate (sum of the bitrate all EMMs). Given in units of kbps.

7 bisstypes (1.3)

7.1 Type Reference

7.1.1 Algorithm

De/Scrambling Algorithm

The DVB-CSA algorithm uses a 64-bit key, while all the AES algorithms use a 128-bit key.

enum

DVB_CSA	DVB-CSA
AES_ATIS_IDSA	AES ATIS IDSA
AES_ECBT_ECB1	AES-ECBT/AES-ECB1 (Alticast)
AES_ECBL_ECB2	AES-ECBL/AES-ECB2 (Latens)
AES_CBCT_CBC1	AES CBCT CBC1
AES_DVB_CISSA	AES DVB-CISSA
AES_IRDETO_CBCT	AES CBCT Irdeto
AES_CBCT_IV	AES CBCT w/specified Initialization Vector

7.1.2 FixedKeyRawDescrambling

Descrambling Method: Fixed Key Raw

struct

algorithm	Algorithm Scrambling algorithm
key	string Scrambling key
iv	optional string Scrambling Initialization vector (only for AES_CBCT_IV)
serviceMappings	list of FixedKeyServiceMapping A list of service mappings.

7.1.3 FixedKeyRawScrambling

Scrambling Method: Fixed Key Raw

struct

algorithm	Algorithm Scrambling algorithm
key	string Scrambling key
iv	optional string Scrambling Initialization vector (only for AES_CBCT_IV)

7.1.4 FixedKeyServiceMapping

Fixed key descrambling for a list of services

struct

key	string Scrambling key
iv	optional string Scrambling Initialization vector (only for AES_CBCT_IV)
serviceIds	list of int A list of services that the key applies to.

8 bissValidate (1.0)

8.1 Type Reference

8.1.1 FloatRange

Float Range

struct

min	float Minimum value of the range
max	float Maximum value of the range

8.1.2 IntRange

Integer Range

struct

min	int Minimum value of the range
max	int Maximum value of the range

8.1.3 Result

Validation result

struct

result	bool True if config is valid and false otherwise.
msg	optional string An error message, which may be given if the result is invalid.

9 bissVersion1 (1.3)

9.1 Type Reference

9.1.1 CsaKeyExpansion

Method of CSA Key Expansion from 48 to 64 bit

enum

CHECKSUM	Session Word is expanded using checksum.
ZERO	Session Word is expanded using zeros.

9.1.2 Descrambling

BISS version 1 Descrambling

variant

mode1	ModelDescrambling BISS version 1, Mode 1.
modeE	ModeEDescrambling BISS version 1, Mode E.

9.1.3 ModelDescrambling

BISS version 1 Descrambling, Mode 1

struct

expansion	CsaKeyExpansion Expansion method.
sessionWord	string Session Word, 48-bit hex string.
serviceMappings	list of ModelServiceMapping A list of service mappings.

9.1.4 ModelScrambling

BISS version 1 Scrambling, Mode 1

struct

expansion	CsaKeyExpansion Expansion method.
sessionWord	string Session Word, 48-bit hex string.

9.1.5 ModelServiceMapping

BISS version 1 Descrambling, Mode 1 for a list of services

struct

sessionWord	string Session Word, 48-bit hex string.
serviceIds	list of int A list of services that the session word applies to.

9.1.6 ModeEDescrambling

BISS version 1 Descrambling, Mode E

struct

activeId	UUID Active session ID.
expansion	CsaKeyExpansion Expansion method.
encSessionWord	UUID Encrypted Session Word, 128-bit. Reference to BissData UUID.
serviceMappings	list of ModeEServiceMapping A list of service mappings.

9.1.7 ModeEScrambling

BISS version 1 Scrambling, Mode E

struct

expansion	CsaKeyExpansion Expansion method.
sessionWord	UUID Session Word, 128-bit. Reference to BissData UUID.

9.1.8 ModeEServiceMapping

BISS version 1 Descrambling, Mode E for a list of services

struct

activeId	UUID Active session ID. Reference to BissData UUID.
encSessionWord	UUID Encrypted Session Word, 128-bit. Reference to BissData UUID.
serviceIds	list of int A list of services that the session ID and encrypted session word applies to.

9.1.9 Scrambling

BISS version 1 Scrambling

variant

mode1	Mode1Scrambling BISS version 1, Mode 1.
modeE	ModeEScrambling BISS version 1, Mode E.

10 bissVersion2 (1.6)

10.1 Type Reference

10.1.1 Descrambling

BISS version 2 Descrambling

variant

mode1	Mode1Descrambling BISS version 2, Mode 1.
modeE	ModeEDescrambling BISS version 2, Mode E.
modeCA	ModeCADescrambling BISS version 2, Mode CA.

10.1.2 Mode1Descrambling

BISS version 2 Descrambling, Mode 1

struct

sessionWord	string Session Word, 128-bit hex string.
serviceMappings	list of Mode1ServiceMapping A list of service mappings.

10.1.3 Mode1Scrambling

BISS version 2 Scrambling, Mode 1

struct

sessionWord	string Session Word, 128-bit hex string.
-------------	--

10.1.4 Mode1ServiceMapping

BISS version 2 Descrambling, Mode 1 for a list of services

struct

sessionWord	string Session Word, 128-bit hex string.
serviceIds	list of int A list of services that the session word applies to.

10.1.5 ModeCADescrambling

BISS version 2 Descrambling, Mode CA

struct

key	optional UUID Private RSA key. If UUID value is given, this references a Biss-Data UUID. If optional is empty, the buried RSA private key is used.
-----	--

10.1.6 ModeCAScrambling

BISS version 2 Scrambling, Mode CA

struct

receivers	UUID Public RSA keys of receivers. Reference to BissData CSV File UUID or BissDataGroup UUID.
esId	string Entitlement session ID (16-bit hex string).
settings	bissCaSettings.BissCaSettings ModeCa settings.

10.1.7 ModeEDescrambling

BISS version 2 Descrambling, Mode E

struct

activeId	UUID Active session ID. Reference to BissData UUID.
encSessionWord	UUID Encrypted Session Word, 128-bit. Reference to BissData UUID.
serviceMappings	list of ModeEServiceMapping A list of service mappings.

10.1.8 ModeEScrambling

BISS version 2 Scrambling, Mode E

struct

sessionWord	UUID Session Word, 128-bit. Reference to BissData UUID.
-------------	---

10.1.9 ModeEServiceMapping

BISS version 2 Descrambling, Mode E for a list of services

struct

activeId	UUID Active session ID. Reference to BissData UUID.
encSessionWord	UUID Encrypted Session Word, 128-bit. Reference to BissData UUID.

serviceIds

list of int

A list of services that the session ID and encrypted session word applies to.

10.1.10 Scrambling

BISS version 2 Scrambling

variant

mode1

Mode1Scrambling

BISS version 2, Mode 1.

modeE

ModeEScrambling

BISS version 2, Mode E.

modeCA

ModeCAscrambling

BISS version 2, Mode CA.

11 component (1.5)

11.1 Overview

Changelog:

1.5

- Added
 - SCTE27_SUB to ComponentType enum and respective mappings.

11.2 Type Reference

11.2.1 AnyComponent

empty **struct**

11.2.2 ComponentGroup

ComponentGroup: Grouping of ComponentType, used for filtering

enum

UNDEFINED
AUDIO
VIDEO
SUBTITLE
DATA

11.2.3 ComponentLanguage

string

11.2.4 ComponentSelector

Component selector.

@param pid	Selects a component using a specific (source) PID.
@param language	Selects a component by a three character language string.
@param type	Selects a component by its type.
@param group	Selects a component by group membership.
@param any	Selects any component.

variant

pid	Pid
language	ComponentLanguage
componentType	ComponentType
componentGroup	ComponentGroup
any	AnyComponent

11.2.5 ComponentType

A ComponentType is typically identified by a stream type (see ISO 13818-1 Table 2-29), often distinguished with another header field such as a descriptor tag.

enum

MPEG_VIDEO	
MPEG_AUDIO	
TTX	
DVBSUB	
ECM	
PCR	
PRIVATE	
AC_3	
H264	
VBI	
AAC	
VC_1	
SCS_EMM	
DPI_CUE	
EAC3	
AIT	
HBBTV_CAROUSEL	
DATA_CAROUSEL	
H265	
S302M	
JPEG2000	
S2038	
DVBTTMLSUB	
JPEG_XS_VIDEO	
SCTE27_SUB	

11.2.6 Pid

int

11.2.7 SubtitleFormat

enum

DVB	
DVB_HARD_OF_HEARING	
EBU	
EBU_HARD_OF_HEARING	

12 inputStatus (1.1)

12.1 Type Reference

12.1.1 FlowId

int

12.1.2 PidStatus

Per PID status.

@param pid	The PID value	
@param serviceId	The associated service ID (as signalled in PAT/PMT)	
@param ccErrors	Accumulated number of CC errors	
@param dataRate	Bitrate [bit/s]	
@param scrambling	Scrambling status of PID	

struct

pid	int
serviceId	optional int
ccErrors	int
dataRate	bigint
scrambling	Scrambling

12.1.3 Scrambling

Scrambling state of PID

@param NONE	No scrambling
@param EVEN	Scrambling with even control word
@param ODD	Scrambling with odd control word

enum

NONE
EVEN
ODD

13 s2xTypes (1.0)

13.1 Type Reference

13.1.1 CodeRate

Code Rate

enum

F11_45	Fraction 11/45
F1_4	Fraction 1/4
F4_15	Fraction 4/15
F13_45	Fraction 13/45
F14_45	Fraction 14/45
F1_3	Fraction 1/3
F2_5	Fraction 2/5
F9_20	Fraction 9/20
F7_15	Fraction 7/15
F1_2	Fraction 1/2
F1_2_L	Fraction 1/2
F8_15	Fraction 8/15
F8_15_L	Fraction 8/15
F11_20	Fraction 11/20
F5_9_L	Fraction 5/9
F26_45	Fraction 26/45
F26_45_L	Fraction 26/45
F3_5	Fraction 3/5
F3_5_L	Fraction 3/5
F28_45	Fraction 28/45
F23_36	Fraction 23/36
F29_45_L	Fraction 29/45
F2_3	Fraction 2/3
F2_3_L	Fraction 2/3
F31_45_L	Fraction 31/45
F25_36	Fraction 25/36
F32_45	Fraction 32/45
F32_45_L	Fraction 32/45
F13_18	Fraction 13/18
F11_15	Fraction 11/15
F11_15_L	Fraction 11/15
F3_4	Fraction 3/4
F7_9	Fraction 7/9
F4_5	Fraction 4/5
F5_6	Fraction 5/6
F77_90	Fraction 77/90
F6_7	Fraction 6/7; only detected Code Rate in Demodulator status, not valid for Modulator
F7_8	Fraction 7/8
F8_9	Fraction 8/9

F9_10	Fraction 9/10
-------	---------------

13.1.2 Constellation

Constellation

enum

QPSK	QPSK
PSK8	PSK8
APSK8	APSK8
APSK16	APSK16
APSK32	APSK32
APSK64	APSK64
APSK128	APSK128
APSK256	APSK256

13.1.3 FrameLength

FEC Frame Length

enum

SHORT	Short frame length
NORMAL	Normal frame length

13.1.4 PIMode

Physical Layer Scrambling Mode

enum

PL_SEQUENCE	Preferred scrambling sequence as defined in ETSI EN 302 307-2
GOLD_SEQUENCE	Gold Sequence Index
PL_ROOT	18-bit Pseudorandom Binary Sequence (PRBS) root

13.1.5 RollOff

Roll-off

All values are allowed in S1, S2 and S2X mode for simplicity.

enum

D0_05	Decimal number 0.05
D0_10	Decimal number 0.10
D0_15	Decimal number 0.15
D0_20	Decimal number 0.20
D0_25	Decimal number 0.25
D0_35	Decimal number 0.35

13.1.6 System

System (DVB Standard)

enum

S1	DVB-S standard
S2	DVB-S2 standard
S2X	DVB-S2X standard

14 s2xValidate (1.0)

14.1 Type Reference

14.1.1 FloatRange

Float Range

struct

min	float Minimum value of the range
max	float Maximum value of the range

14.1.2 IntRange

Integer Range

struct

min	int Minimum value of the range
max	int Maximum value of the range

14.1.3 Result

Validation result

struct

result	bool True if config is valid and false otherwise.
msg	optional string An error message, which may be given if the result is invalid.

15 s2xinCardSettings (1.0)

15.1 Command Reference

15.1.1 GetCardSetting

- message **GetCardSetting.Request**
- message **GetCardSetting.Response**

15.1.2 SetCardSetting

- message **SetCardSetting.Request**
- message **SetCardSetting.Response**

15.2 Type Reference

15.2.1 CardSettings

Settings for the operational mode of the card.

struct

mode	DemodulatorMode Number of available demodulators
------	--

15.2.2 DemodulatorMode

Number of available demodulators

enum

Demods4	4 demodulators available
Demods8	8 demodulators available
Demods16	16 demodulators available
Demods32	32 demodulators available

15.2.3 GetCardSetting.Request

empty **struct**

15.2.4 GetCardSetting.Response

struct

data	map from UUID to CardSettings
------	--

15.2.5 SetCardSetting.Request

struct

data	map from UUID to CardSettings
------	--

15.2.6 SetCardSetting.Response

empty **struct**

16 s2xinDemodulator (1.8)

16.1 Overview

1.10

- **Changed**
 - the imported package bissDescrambling from 1.5 to 1.6;

1.7

- **Changed**
 - the imported package bissDescrambling from 1.4 to 1.5;

1.6

- **Changed**
 - the imported package bissDescrambling from 1.3 to 1.4;

16.2 Command Reference

16.2.1 GetDemodulators

Get Demodulator objects from database.

- message **GetDemodulators.Request**
- message **GetDemodulators.Response**

16.2.2 SetDemodulators

Update Demodulator objects in database.

Note: This interface should *NOT* be used to create any new Demodulator objects. Instead, use it only to modify existing objects.

- message **SetDemodulators.Request**
- message **SetDemodulators.Response**

16.2.3 DeleteDemodulators

Delete Demodulator objects from database.

Note: This command should *NOT* be used (except for debugging purposes). Demodulator objects should never be deleted through the API.

- message **DeleteDemodulators.Request**
- message **DeleteDemodulators.Response**

16.2.4 CheckUsageInputs

Check if a set of inputs are in use by any outputs

- message **CheckUsageInputs.Request**
- message **CheckUsageInputs.Response**

16.3 Type Reference

16.3.1 AnalyzeMode

Analyze Mode tells the system what type of input is expected, hence what kind of analysis may be performed on the incoming stream.

enum

dvbMode	Analyze the input as DVB TS
mpegMode	Analyze the input as MPEG TS

16.3.2 CheckUsageInputs.Request

CheckUsageInputsRequest

16.3.3 CheckUsageInputs.Response

CheckUsageInputsResponse

16.3.4 CheckUsageInputsRequest

CheckUsageInputs request

struct

ids	list of UUID
-----	--------------

16.3.5 CheckUsageInputsResponse

Response of a CheckUsageInputs request

struct

data	map from UUID to bool
------	-----------------------

16.3.6 DeleteDemodulators.Request

struct

ids	list of UUID
-----	--------------

16.3.7 DeleteDemodulators.Response

empty struct

16.3.8 Demodulator

Demodulator Configuration

struct

label	string User label for the transport stream
enabled	bool Enable / disable demodulator
port	s2xinPhysicalPort.Port Physical input port
satelliteFrequency	float Satellite frequency (in MHz). The sum of LNB frequency and satellite frequency is used to derive the actual L-band frequency on the input connector.
streamIndex	optional int ISI (Input Stream Identifier). Setting ISI number is only applicable if the incoming stream is MIS (Multiple Input Stream).
symbolRate	float Symbol rate (in Mbd)
plScramblingMode	s2xTypes.PIMode Physical Layer scrambling mode
plScramblingCode	int Physical Layer scrambling code sequence, in unit according to plScramblingMode
analyzeMode	AnalyzeMode Analyze properties for this input
descrambling	bissDescrambling.DescramblerSettings Descrambler settings

16.3.9 GetDemodulators.Request

empty **struct**

16.3.10 GetDemodulators.Response

struct

data **map** from **UUID** to **Demodulator**

16.3.11 SetDemodulators.Request

struct

data **map** from **UUID** to **Demodulator**

16.3.12 SetDemodulators.Response

empty **struct**

17 s2xinPhysicalPort (1.0)

17.1 Command Reference

17.1.1 GetPhysicalPorts

Get PhysicalPort objects from database.

- message **GetPhysicalPorts.Request**
- message **GetPhysicalPorts.Response**

17.1.2 SetPhysicalPorts

Update PhysicalPort objects in database.

Note: This interface should *NOT* be used to create any new PhysicalPort objects. Instead, only use it to modify existing objects.

- message **SetPhysicalPorts.Request**
- message **SetPhysicalPorts.Response**

17.1.3 DeletePhysicalPorts

Delete PhysicalPort objects from database.

Note: This command should *NOT* be used (except for debugging purposes). PhysicalPort objects should never be deleted through the API.

- message **DeletePhysicalPorts.Request**
- message **DeletePhysicalPorts.Response**

17.2 Type Reference

17.2.1 DeletePhysicalPorts.Request

struct

ids [list of UUID](#)

17.2.2 DeletePhysicalPorts.Response

empty **struct**

17.2.3 GetPhysicalPorts.Request

empty **struct**

17.2.4 GetPhysicalPorts.Response

struct

data [map from UUID to PhysicalPort](#)

17.2.5 LnbControl

LNB Control

struct

dc	LnbDc Direct current output voltage from LNB
tone22kHz	bool Enable/Disable 22kHz tone output from LNB The 22 kHz tone can not be enabled if DC is set to 0V.

17.2.6 LnbDc

LNB Direct Current Output

enum

DC_0V	Output 0V DC
DC_13V	Output 13V DC
DC_18V	Output 18V DC

17.2.7 PhysicalPort

Physcial port

Settings related to each of the physical ports.

struct

port	Port Physical input port
label	string User defined port name
lnbFrequency	float LNB frequency (in MHz)
lnbControl	LnbControl LnbControl

17.2.8 Port

Physical Input Port

The ports are numbered from left to right on the card. Each demodulator can select a transponder from 2 physical input ports. The first half of the demodulators can select P1 or P2, while the second half can select P3 or P4.

enum

P1	Port 1
P2	Port 2
P3	Port 3
P4	Port 4

17.2.9 SetPhysicalPorts.Request

struct

data **map** from **UUID** to **PhysicalPort**

17.2.10 SetPhysicalPorts.Response

empty **struct**

18 s2xinStatus (1.6)

18.1 Overview

1.6

- **Changed**
 - the imported package component from 1.4 to 1.5;
 - the imported package tsstatus from 1.3 to 1.5;

1.5

- **Changed**
 - the imported package component from 1.1 to 1.4;
 - the imported package tsstatus from 1.1 to 1.3;

18.2 Command Reference

18.2.1 GetDemodulatorStatus

Get Demodulator status

- message **GetDemodulatorStatus.Request**
- message **GetDemodulatorStatus.Response**

18.2.2 GetTransportStreamStatus

Get TransportStream status

- message **GetTransportStreamStatus.Request**
- message **GetTransportStreamStatus.Response**

18.2.3 ClearAllCounters

Clear all Demodulator Status Counters

This resets the CC error count.

- message **ClearAllCounters.Request**
- message **ClearAllCounters.Response**

18.3 Type Reference

18.3.1 ClearAllCounters.Request

empty **struct**

18.3.2 ClearAllCounters.Response

empty **struct**

18.3.3 CodingModulation

Coding Modulation

enum

ACM	Adaptive Coding and Modulation (ACM)
CCM	Constant Coding and Modulation (CCM)

18.3.4 DemodulatorStatus

Demodulator status

struct

source	UUID The source on this demodulator
locked	bool Lock indicator, if true signalStatus struct should be filled
signalStatus	optional SignalStatus Signal status
fpgaStatus	FpgaStatus FPGA status
index	int Internal index (used for debugging)

18.3.5 DemodulatorStatusRequest

Demodulator status request

struct

ids	optional list of UUID
-----	------------------------------

18.3.6 DemodulatorStatusResponse

Response of a Demodulator status request

struct

data	map from UUID to DemodulatorStatus
------	---

18.3.7 FpgaStatus

FPGA status

struct

totalBitrate	float Total Bitrate (in Mbps)
effectiveBitrate	float Effective Bitrate (in Mbps)

18.3.8 GetDemodulatorStatus.Request

DemodulatorStatusRequest

18.3.9 GetDemodulatorStatus.Response

DemodulatorStatusResponse

18.3.10 GetTransportStreamStatus.Request

TransportStreamStatusRequest

18.3.11 GetTransportStreamStatus.Response

TransportStreamStatusResponse

18.3.12 SignalStatus

Signal status

struct

actualFrequency	float Found frequency (in MHz)
carrierOffset	float Offset of the frequency (in MHz)
actualSymbolRate	float Found symbol rate (MBd)
symbolRateOffset	float Offset of the symbol rate (MBd)
carrierNoiseRatio	float Carrier to noise ratio (in dB)
linkMargin	optional float Margin in between required and measured carrier to noise ratio (in dB)
receivedPackets	bigint Total number of recieved packets since lock
errorPackets	bigint Total number of error packets since lock
packetErrorRate	float Packet Error Rate
ebNo	optional float Energy per bit to noise power spectral density ratio (in dB/Hz)
inputPower	float Input Power (in dBm)
rolloff	optional s2xTypes.Rolloff Roll-off factor (between 0.05 and 0.35)
modulationStandard	optional s2xTypes.System DVBS or DVBS2
constellation	optional s2xTypes.Constellation Constellation

codeRate	optional s2xTypes.CodeRate Code Rate
pilots	bool True if pilots are enabled
frameLength	optional s2xTypes.FrameLength Short or Normal
inverted	bool Inversion of the signal, true if inverted
isi	optional int ISI (Input Stream Identifier). If set, MIS (Multiple Input Stream), else SIS (Single Input Stream)
issyi	bool Input Stream Synchronization Indicator active/not active (true/false). If active, the ISSY data is inserted after each TS packet
codingModulation	optional CodingModulation Constant Coding and Modulation (ACM) or Adaptive Coding and Modulation (CCM)
nullPacketDeletion	bool Null-packet deletion active/not active (true/false)

18.3.13 TransportStreamStatusRequest

TransportStream status request

struct

ids **optional list of UUID**

18.3.14 TransportStreamStatusResponse

Response of a TransportStream status request

struct

data **map from UUID to tsstatus.TransportStreamStatus**

19 tsstatus (1.5)

19.1 Overview

Changelog:

1.5

- Added
 - RPC `GetOutputPidStatus` was added, which takes in a list of uuid of outputs and returns the status of the respective output transport streams.

1.4

- Changed
 - component from 1.4 to 1.5

1.3

- Changed
 - Dependency module component bumped from version 1.1 to 1.4

19.2 Command Reference

19.2.1 `GetOutputPidStatus`

Get output PID status

- message `GetOutputPidStatus.Request`
- message `GetOutputPidStatus.Response`

19.3 Type Reference

19.3.1 `GetOutputPidStatus.Request`

`OutputPidStatusRequest`

19.3.2 `GetOutputPidStatus.Response`

`OutputPidStatusResponse`

19.3.3 `OutputPidStatusRequest`

Output PID status request

`struct`

outputs `list of UUID`

19.3.4 `OutputPidStatusResponse`

Response of an output PID status request

`struct`

data

map from **UUID** to **OutputTransportStreamStatus**

19.3.5 OutputTransportStreamStatus

Status for the output TS

struct

services

map from **int** to **ServiceSourceStatus**

Status for each service in the TS. Map from service ID to status struct.

pids

list of **PidStatus**

Info on all PIDs, both signalled and unsignalled. Map from service PID value to status struct.

19.3.6 PidStatus

PidStatus state of PID

struct

serviceId

list of **int**

List of serviceIds a PID belong to.

componentType

optional **component.ComponentType**

ComponentType of PID

ccErrors

int

CC errors on PID

dataRate

bigint

Data rate of PID

scrambling

Scrambling

Scrambling of PID (none, even or odd)

PCR

bool

present

bool

True if signalled pid has rate

language

optional string

Language of PID, if applicable (e.g. audio pid)

casId

optional int

Id of CAS system, if applicable (e.g. ECM pid)

pidIn

int

PID number on input

pidOut

optional int

PID number on output

19.3.7 Scrambling

Scrambling state of PID

enum

NONE

No scrambling

EVEN

Scrambling with even control word

ODD

Scrambling with odd control word

19.3.8 ServiceSourceStatus

Status for an input service

struct

serviceName	optional string Service name
pmtPid	int PMT pid for service
pcrPid	optional int PCR pid for service if present

19.3.9 TransportStreamStatus

Status for the TS

struct

ccErrors	int Total CC error count for this TS.
services	map from int to ServiceSourceStatus Status for each service in the TS. Map from service ID to status struct.
pids	map from int to PidStatus Info on all PIDs, both signalled and unsignalled. Map from service PID value to status struct.
tsId	optional int Transport Stream ID; set if PAT pid present in stream
originalNetworkId	optional int Original Network ID; set if SDT pid present in stream